

36 — BAYESIAN BELIEF NETWORK USING PGMPY

Aim

To construct a Bayesian Belief Network and perform probabilistic inference using the pgmpy library.

Software Required

- Python 3.x
- Jupyter Notebook / Google Colab
- pgmpy

Installation

```
!pip install pgmpy
```

Algorithm + Procedure

1. Define the Bayesian Network structure.
2. Create nodes for Rain, Sprinkler, and Wet.
3. Define the prior probability distributions.
4. Define the conditional probability distribution of Wet.
5. Add the CPDs to the network.
6. Verify the network.
7. Create a Variable Elimination object.
8. Provide Wet as observed evidence.
9. Calculate the posterior probability of Rain.
10. Display the inference result.

Code

```
from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.factors.discrete import TabularCPD
from pgmpy.inference import VariableElimination
model = DiscreteBayesianNetwork([
    ('Rain', 'Wet'),
    ('Sprinkler', 'Wet')
])
```

```

cpd_rain = TabularCPD(
    'Rain',
    2,
    [[0.8],
     [0.2]]
)

cpd_sprinkler = TabularCPD(
    'Sprinkler',
    2,
    [[0.6],
     [0.4]]
)

cpd_wet = TabularCPD(
    'Wet',
    2,
    [[1.0, 0.1, 0.1, 0.01],
     [0.0, 0.9, 0.9, 0.99]],
    evidence=['Rain', 'Sprinkler'],
    evidence_card=[2, 2]
)

model.add_cpds(
    cpd_rain,
    cpd_sprinkler,
    cpd_wet
)

print("Model valid:", model.check_model())

infer = VariableElimination(model)

result = infer.query(
    variables=['Rain'],

```

```
evidence={'Wet': 1}  
  
)  
  
print(result)
```

Output

```
/usr/local/lib/python3.13/dist-packa  
from .StructureScore import (  
Model valid: True  
+-----+-----+  
| Rain   | phi(Rain) |  
+=====+=====+  
| Rain(0) | 0.6061 |  
+-----+-----+  
| Rain(1) | 0.3939 |  
+-----+-----+
```